1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlights a tension where tumor-intrinsic immunogenicity from MMR deficiency (MSH6) should sensitize tumors to immunotherapy, yet adipocyte-rich, stromal-remodeled tumors (SARIFA) and systemic lipid/metabolic states (HDL-C) may promote immune exclusion and metastatic behavior—why do some MSH6-deficient MIBC tumors fail to realize an immunogenic benefit?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), MSH6 deficiency increases tumor neoantigen burden and baseline immunogenicity, but the observed immunotherapy response is critically shaped by the stromal–lipid microenvironment. Specifically, TSR-low tumors will permit enhanced T-cell infiltration and superior response to immune checkpoint blockade in the context of MSH6 deficiency, whereas TSR-high, adipocyte-rich stroma fosters a lipid-rich, TAF-mediated immune-excluding niche that blunts neoantigen presentation and T-cell infiltration, thereby diminishing therapy_response despite MSH6 deficiency. Elevated HDL-C is predicted to amplify this lipid-mediated immune exclusion, further reducing therapy_response in the TSR-high/MSH6-deficient subset. Therefore, the highest therapy_response is expected in tumors that are MSH6-deficient with low TSR and low HDL-C, and the lowest in tumors that are MSH6-deficient with high TSR and high HDL-C; intermediate outcomes are anticipated for other combinations. This hypothesis is testable by (1) measuring MSH6 status, TSR, and HDL-C in MIBC patients treated with ICIs and assessing therapy_response; (2) modeling interaction effects (MSH6 × TSR × HDL-C) on response, with adjustments for known confounders; and (3) validating mechanisms in relevant preclinical models (organoid/CAF co-cultures and MSH6-deficient mouse bladder cancer with diet-induced adiposity).

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	MSH6, HDL-C
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6) should sensitize tumors to immunotherapy, yet adipocyte-rich, stromal-remodeled tumors (SARIFA) and systemic lipid/metabolic states (HDL-C) may promote immune exclusion and metastatic behavior—why do some MSH6-deficient MIBC tumors fail to realize an immunogenic benefit? (2) Tier 1 paths used: I built on MSH6 (MMR/neoantigen axis) and SARIFA/HDL-C (microenvironmental stromal–adipocyte phenotype and systemic lipid milieu) because they form a coherent mechanistic triad linking neoantigen load, local immune accessibility, and lipid-mediated immune suppression.

HDL-C: HDL-C) may promote immune exclusion and metastatic behavior—why do some MSH6-deficient MIBC tumors fail to realize an immunogenic benefit? (2) Tier 1 paths used: I built on MSH6 (MMR/neoantigen axis) and SARIFA/HDL-C (microenvironmental stromal–adipocyte phenotype and systemic lipid milieu) because they form a coherent mechanistic triad linking neoantigen load, local immune accessibility, and lipid-mediated immune suppression.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	PTGDS	Supporting modifier from knowledge graph
Secondary	KRT17	Supporting modifier from knowledge graph
Secondary	MDK	Supporting modifier from knowledge graph
Secondary	EV_mRNA_markers	Supporting modifier from knowledge graph
Secondary	MSI	Supporting modifier from knowledge graph
Secondary	TAF	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	GLU_urinary_creatinine_microalbuminuria	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PTGDS: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- KRT17: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.